



Practical Exam

Software Development Life Cycle and Application

Names:	Garcia, Dan Matthew P. Orteza, Clayborne Benedict	Date:	September 26, 2026
Program:	BS CPE		

A. Scenario:

A local organization is looking for a simple system to automate one of its manual processes. Your group must identify a real-world problem and develop a software solution with the system you have picked from the roster.

B. Planning and Requirements Analysis:

System Proposal

Requirement Gathering

- Identify function and non-functional requirements

Development

- Minimum features
 - Search Functions
 - Data Validation
 - User-Friendly Interface
- Other features
 - Applicant Management
 - Register new applicants with full personal profile (name, contact info, DOB, gender, address)
 - Auto-generated unique applicant IDs
 - View list of all registered applicants
 - Academic & Financial Profiling
 - Capture academic background per application (year level, program, school, school type, GWA, TOR/Form 138 path)
 - Capture financial/socio-economic profile (household income, parent/guardian info, employment status, supporting document paths)
 - Scholarship Program Management
 - 9 real Mapúa scholarship programs hardcoded with actual eligibility criteria, benefits, and required documents
 - Register additional/custom scholarship programs
 - View full program details (benefits, eligibility, requirements, responsibilities, application period)
 - Application Lifecycle Management
 - Create applications linking an applicant to a scholarship program
 - 5-state status tracking (Draft, Submitted, Under Evaluation, Approved, Rejected)
 - Manual status updates and evaluator score logging
 - Edit-in-place: select an existing application to update its details instead of creating a duplicate



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- Eligibility Matching
 - Automated check against year level, program/field, income cap, and GWA cap per scholarship
 - Reports specific unmet criteria per program, not just a pass/fail
 - Documented limitation: SHS percentage-scale criteria vs. college 1.0–5.0 GWA scale are not numerically comparable, flagged as informational-only
- Search Functions
 - Search by applicant name, applicant ID, status, scholarship type, or program name
 - Filter by GWA range and by evaluation score range
 - Combinable advanced search across multiple filters at once
- Data Validation
 - Name, email, and PH mobile number format checks
 - Date of birth, GWA (1.0–5.0), and income validation
 - File upload validation (extension + existence)
 - Required-field enforcement
 - Evaluation score range validation
 - Pre-submission completeness check before Draft → Submitted
- User-Friendly Interface
 - Tkinter GUI with Mapúa-branded color scheme (maroon/gold/black/grey)
 - Tabbed layout: Applicants, Scholarship Programs, Applications, Search
 - Dropdowns/comboboxes for fixed-choice fields instead of free typing
 - Scrollable tabs so no content or button is ever hidden off-screen

Testing

- Notify your instructor when it is time to perform testing.
- Minimum
 - 3 use cases
 - UI
 - Eligibility
 - Input of Score
 - Search
 - Categorized

Challenges and Limitations



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C. Submission Requirements

- Project Documentation (PDF)
- Source Code
- Selfie

OUTPUT UI

Scholarship Application System
Applicant records, scholarship grants, and eligibility matching.

Applicants | Scholarship Programs | Applications | Search

Register New Applicant

First Name* Middle Name

Last Name* Suffix (jr, III)

Email* Mobile (9999999999)*

Date of Birth (YYYY-MM-DD)* Permanent Address*

Gender*

Registered Applicants

Applicant ID	Full Name	Email	Mobile
APP-00001	Domingo, Kristine Randa	kristine.domingo@example.com	09091234503
APP-00002	Calillo, Andrea Lim	andrea.calillo@example.com	09081234502
APP-00003	Reyes, Miguel Santos	miguel.reyes@example.com	09111234501

Applicants: 3 | Programs: 8 | Applications: 5 | Tip: scroll within a tab if content runs below the window

Scholarship Application System
Applicant records, scholarship grants, and eligibility matching.

Applicants | Scholarship Programs | Applications | Search

Scholarship Programs

Program ID	Grant Name	Provider	Type
PRG-002	Alfonso T. Yuchengco (ATY) Scholarship	Alfonso T. Yuchengco Foundation / Mapúa CSFA	Financial Need-Based
PRG-003	Don Tomas Mapúa Scholarship	Mapúa University / CSFA	Academic Merit
PRG-004	Academic Scholarship	Mapúa University / CSFA	Academic Merit
PRG-005	Mapúa Alumni Australia (MAA) Scholarship	Mapúa Alumni Australia	Financial Need-Based
PRG-006	Mapúa Alumni Association - Alberta Chapter (MAAAC) Scholarship	Mapúa Alumni Association - Alberta Chapter	Academic Merit
PRG-007	Mapúa Alumni Association of San Diego (MAASD) Scholarship	Mapúa Alumni Association of San Diego	Academic Merit
PRG-008	NET CE-Ense Alumni Association, Inc. Scholarship	NET CE-Ense Alumni Association, Inc.	Academic Merit
PRG-009	Southern California Mapúa Alumni (SCMA) Scholarship	Southern California Mapúa Alumni	Academic Merit

Program Details (select a row above)

Southern California Mapúa Alumni (SCMA) Scholarship (Southern California Mapúa Alumni)
Type: Academic Merit
Total modeled coverage: Php 400.00

Benefits:

- \$400 grant per term

Eligibility criteria:

- Incoming 3rd or 4th Year Engineering or Architecture student
- GWA standing of 2.50 for the last two terms

Register New Scholarship Program

Grant Name* Provider* Type*

Applicants: 3 | Programs: 8 | Applications: 5 | Tip: scroll within a tab if content runs below the window



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Scholarship Application System
Applicant records, scholarship grants, and eligibility matching

Applicants | **Scholarship Programs** | Applications | Search

Scholarship Programs

Program ID	Grant Name	Provider	Type
PRG-002	Alfonso T. Yachengco (ATY) Scholarship	Alfonso T. Yachengco Foundation / Mapúa CSEA	Financial Need Based
PRG-003	Don Tomas Mapúa Scholarship	Mapúa University / CSEA	Academic Merit
PRG-004	Academic Scholarship	Mapúa University / CSEA	Academic Merit
PRG-005	Mapúa Alumni Australia (MAA) Scholarship	Mapúa Alumni Australia	Financial Need Based
PRG-006	Mapúa Alumni Association - Alberta Chapter (MAAAC) Scholarship	Mapúa Alumni Association - Alberta Chapter	Academic Merit
PRG-007	Mapúa Alumni Association of San Diego (MAASD) Scholarship	Mapúa Alumni Association of San Diego	Academic Merit
PRG-008	MIT CE-Ense Alumni Association, Inc. Scholarship	MIT CE-Ense Alumni Association, Inc.	Academic Merit
PRG-009	Southern California Mapúa Alumni (SCMA) Scholarship	Southern California Mapúa Alumni	Academic Merit

Program Details (select a row above)

MIT CE-Ense Alumni Association, Inc. Scholarship (MIT CE-Ense Alumni Association, Inc.)
Type: Academic Merit
Total awarded coverage: Php 0.00

Benefits:

- Full matriculation fee

Eligibility criteria:

- Must be currently enrolled as a BS Civil Engineering or BS Environmental and Sanitary Engineering student
- Must be at least 3rd year standing

Register New Scholarship Program

Grant Name* Provider* Type*

Applications: 5 | Programs: 9 | Applications: 3 | Tip: scroll within a list if content runs below the window

Scholarship Application System
Applicant records, scholarship grants, and eligibility matching

Applicants | Scholarship Programs | **Applications** | Search

Year/Grade Level* Enrolled Program/Thread*

SSN Year BS Architecture

School Name* GWA (1.0-5.0)*

Mapúa University 1.8

TOR/Form 138 file path* Household Monthly Income*

Father's Name 4000.0

Edgar Domingo Father's Occupation

Driver

Editing SCH-0001

Applications (select a row to edit or act on it)

Application ID	Applicant	Program	Status	Score
SCH-0001	Domingo, Kristine Pineda	Mapúa Alumni Australia (MAA) Scholarship	Draft / Incomplete	
SCH-0002	Cadillo, Andrea Lim	Mapúa Alumni Association - Alberta Chapter (MAAAC) Scholar	Draft / Incomplete	
SCH-0003	Rivera, Miguel Santos	Southern California Mapúa Alumni (SCMA) Scholarship	Draft / Incomplete	

Applications: 5 | Programs: 9 | Applications: 3 | Tip: scroll within a list if content runs below the window



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Scholarship Application System

Scholarship Application System

Applicant records, scholarship grants, and eligibility checking

Applicants

Year/Grade Level

4th Year

School Name

Mapúa University

TOR Form 128

Father's Name

Edgar Domingo

Listing SCN#

00001

Enrolled Program/Strand

BS Architecture

GWA (1.0-5.0)

1.8

Household Monthly Income

40000

Father's Occupation

Driver

Clear Form / New Application

Update Application

Applications

Application ID

Applicant

Program

Status

Score

SCH-00001

Domingo, Edwin Feado

Mapúa Alumni Australia (MAA) Scholarship

Draft / incomplete

99.0

SCH-00002

Castro, Andres Lim

Mapúa Alumni Association - Alberta Chapter (MAAAC) Scholar

Draft / incomplete

SCH-00003

Royes, Miguel Santos

Southern California Mapúa Alumni (SCMA) Scholarship

Draft / incomplete

Submit Application

Update Status

Log Evaluation Score

Check Eligibility

Scholarship Application System

Applicant records, scholarship grants, and eligibility checking

Applicants

Scholarship Programs

Applications

Search

Year/Grade Level

4th Year

School Name

Mapúa University

TOR Form 128 file path

Father's Name

Edgar Domingo

Listing SCN#

00001

Enrolled Program/Strand

BS Architecture

GWA (1.0-5.0)

1.8

Household Monthly Income

40000

Father's Occupation

Driver

Clear Form / New Application

Update Application

Applications (select a row to edit or act on it)

Application ID

Applicant

Program

Status

Score

SCH-00001

Domingo, Edwin Feado

Mapúa Alumni Australia (MAA) Scholarship

Draft / incomplete

99.0

SCH-00002

Castro, Andres Lim

Mapúa Alumni Association - Alberta Chapter (MAAAC) Scholar

Draft / incomplete

SCH-00003

Royes, Miguel Santos

Southern California Mapúa Alumni (SCMA) Scholarship

Draft / incomplete

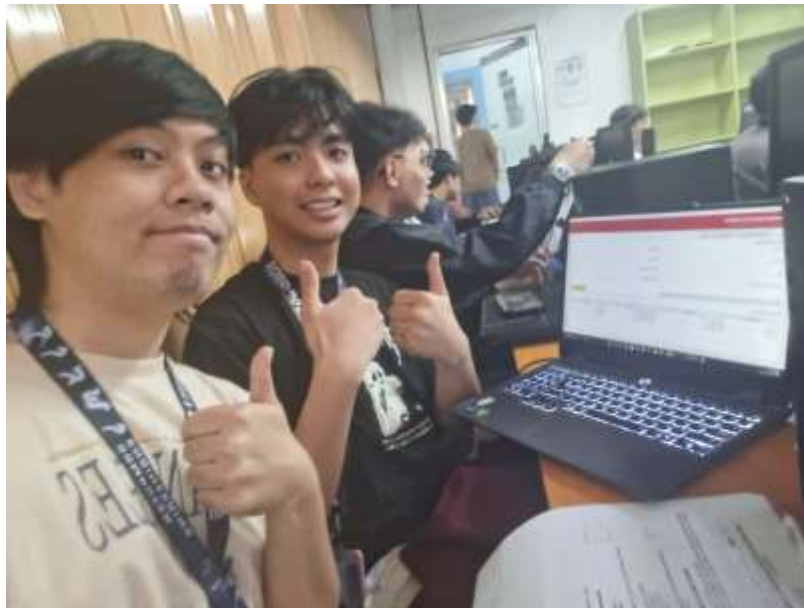
Submit Application

Update Status

Log Evaluation Score

Check Eligibility

Software Development Life Cycle and Application
PICTURE/PROOF OF DOING THE ACTIVITY





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Software Development Life Cycle and Application

E. Score Sheet

Criteria	Poor (25%)	Fair (50%)	Good (75%)	Excellent (100%)	Score
I. Completeness, Documentation, and Organization of Report	The report was incomplete, messy, and had many unanswered questions.	The report was complete, messy, and had many unanswered questions.	The report was complete, neat, and had 1 or 2 unanswered questions.	The report was complete, neat, and all the questions had been answered.	
II. Coding Design and Patterns	The program had inappropriate elements and structures, incorrect indentation, and poor program documentation.	The program had corrected indention, but inappropriate elements and structures and poor program documentation.	The program had corrected indention, adequate program documentation, but inappropriate elements and structures.	The program had appropriate elements and structures, correct indention, and excellent program documentation.	
III. Findings, Observations, and Comments	The findings. observations, and comments were not based on the gathered data and results. All thoughts were inconsistent and unclear.	The findings. observations, and comments based on the gathered data and results but were not elaborated. Not all thoughts were consistent and clear.	The findings. observations, and comments were based on the gathered data and results and were mostly elaborated. Most of the thoughts were consistent and clear.	The findings. observations, and comments were based on the gathered data and results and were completely elaborated. All the thoughts were consistent and clear.	
IV. Grammar and Composition	The words used were inappropriate, wrong grammar usage, and poor sentence construction was observed.	The words used were appropriate; however, bad grammar and poor sentence construction were observed.	The words used were appropriate, had proper grammar usage, but poor sentence construction was observed.	The words used were appropriate, had proper grammar usage, and excellent sentence construction.	
SCORE:					

CHECKED			
Rating		Signature	
		Date	